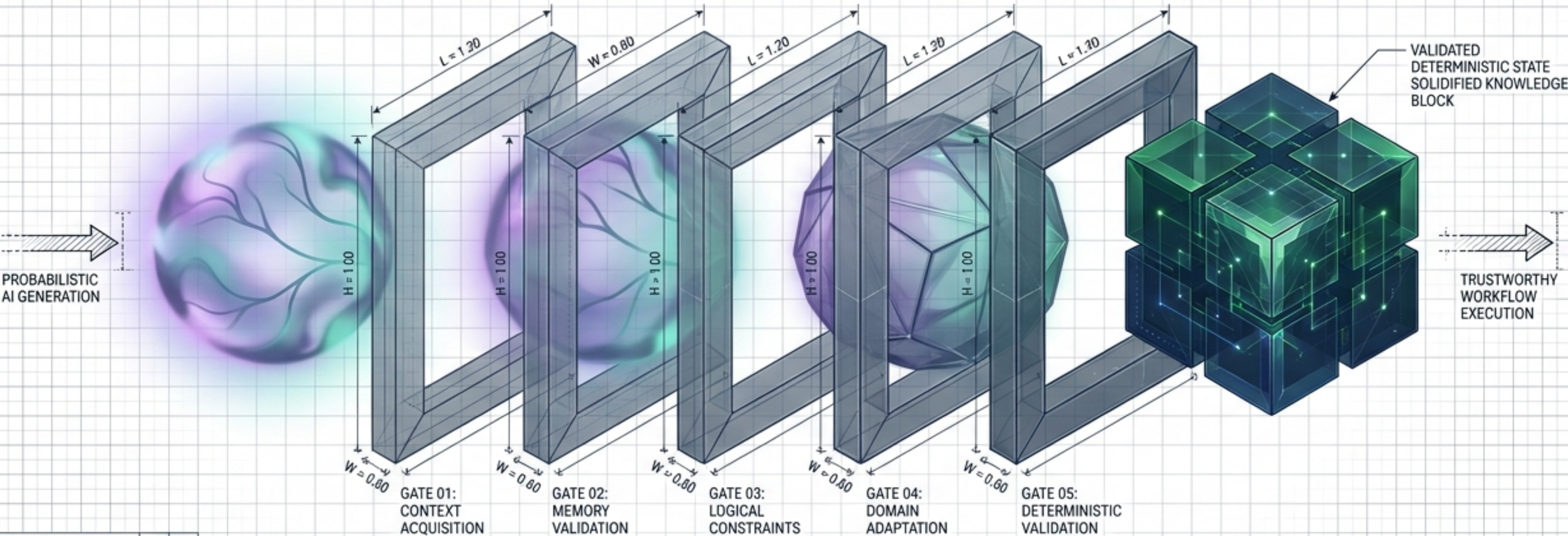


REV	REASON	DATE

ACME

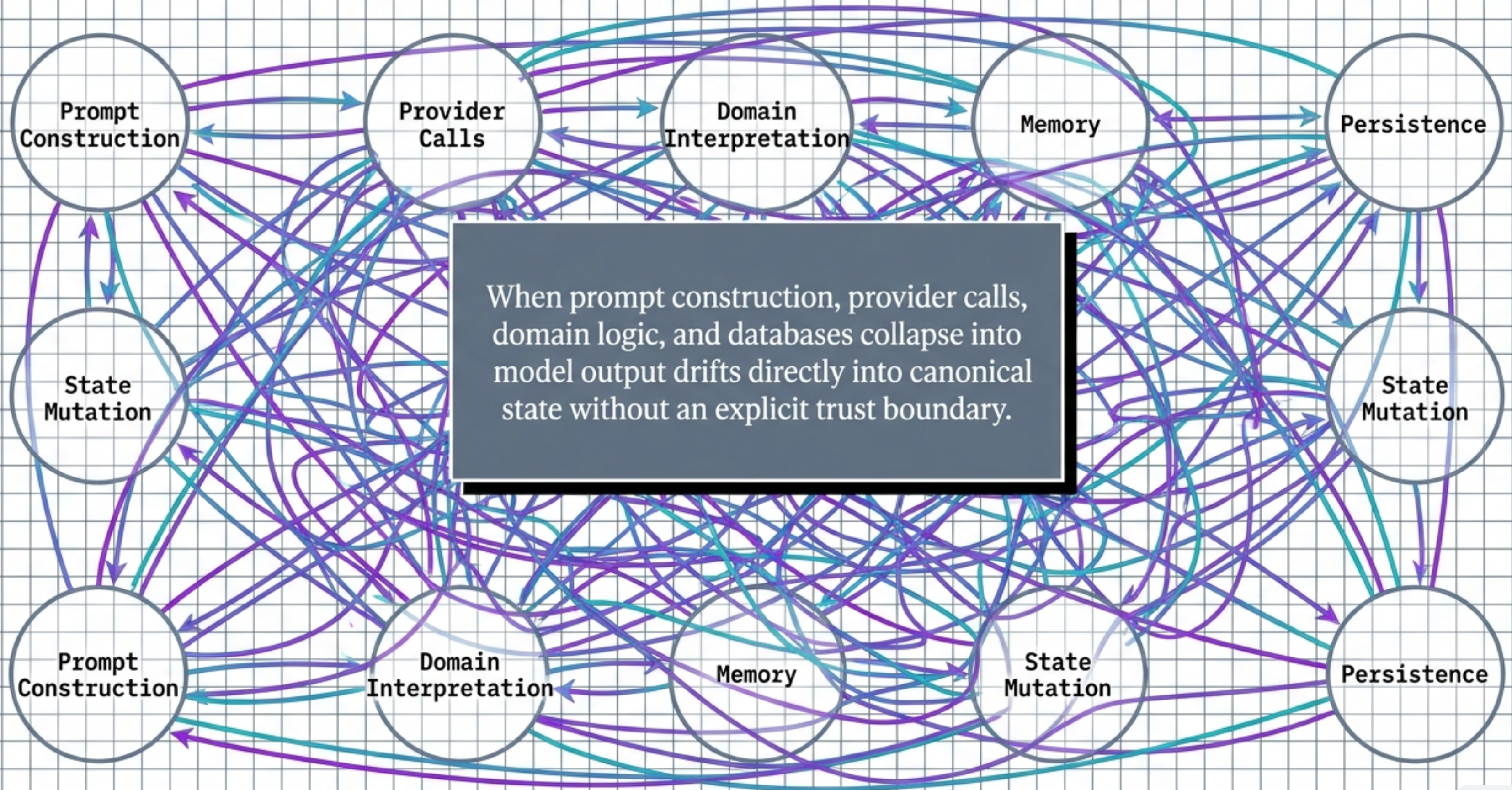
Adaptive Context Memory Engine

A domain-neutral, replayable execution system for trustworthy AI workflows.



DATE	REVISION	REV.	REV.

The risk starts when one service owns everything

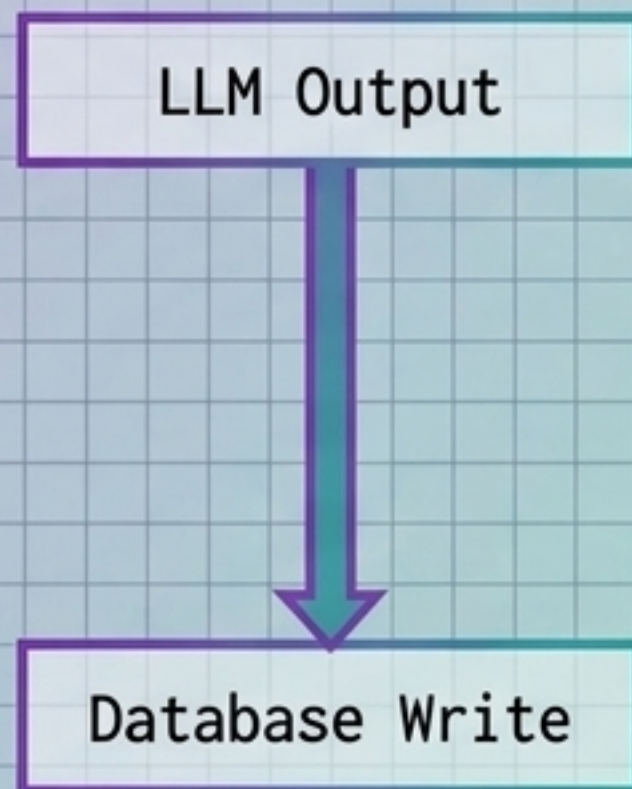


Model output is a candidate, never truth



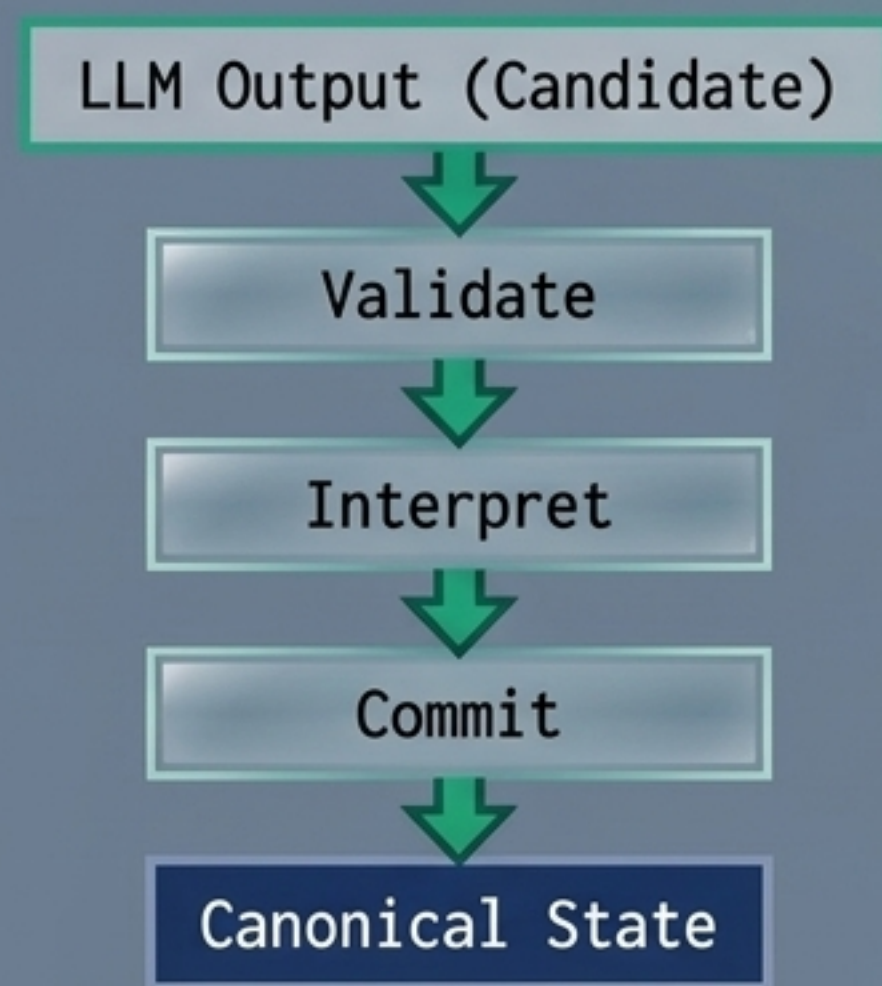
Silent Overwrite

Probabilistic models act as truth

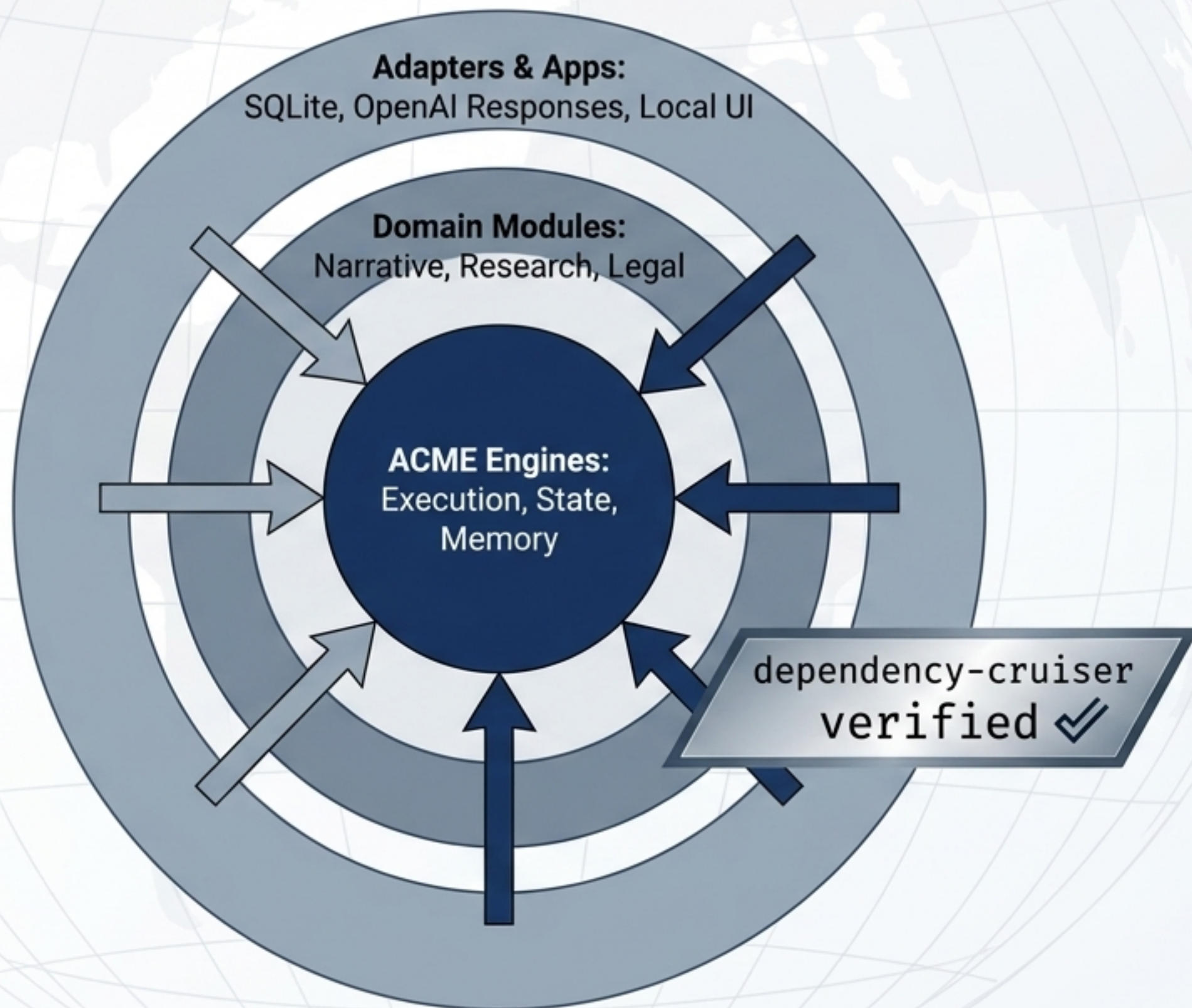


Immutable Audit

Deterministic contracts, policies, and ledgers decide



The Glass Ledger: Ports, Adapters, and the Neutral Core

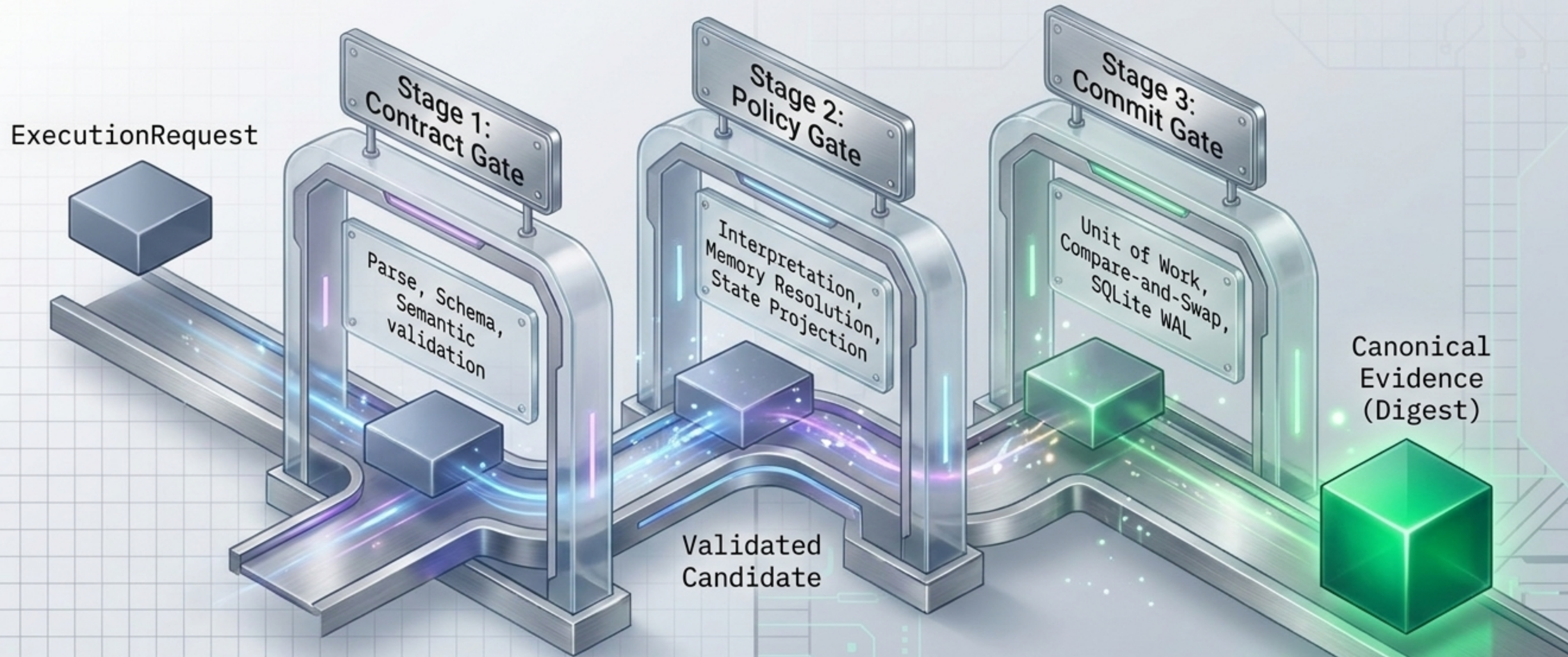


The core knows
nothing of domains.

Modules know
nothing of adapters.

Dependencies point
strictly inward toward
public core
contracts.

One execution crosses three strict trust stages



Separating how memory works from what memory means



Core MemoryEngine

Correlation

Ordering

Versioned Mutation

Timestamps

Working Sets



Domain Module

Equivalence

Contradiction

Promotion Rules

Confidence Thresholds

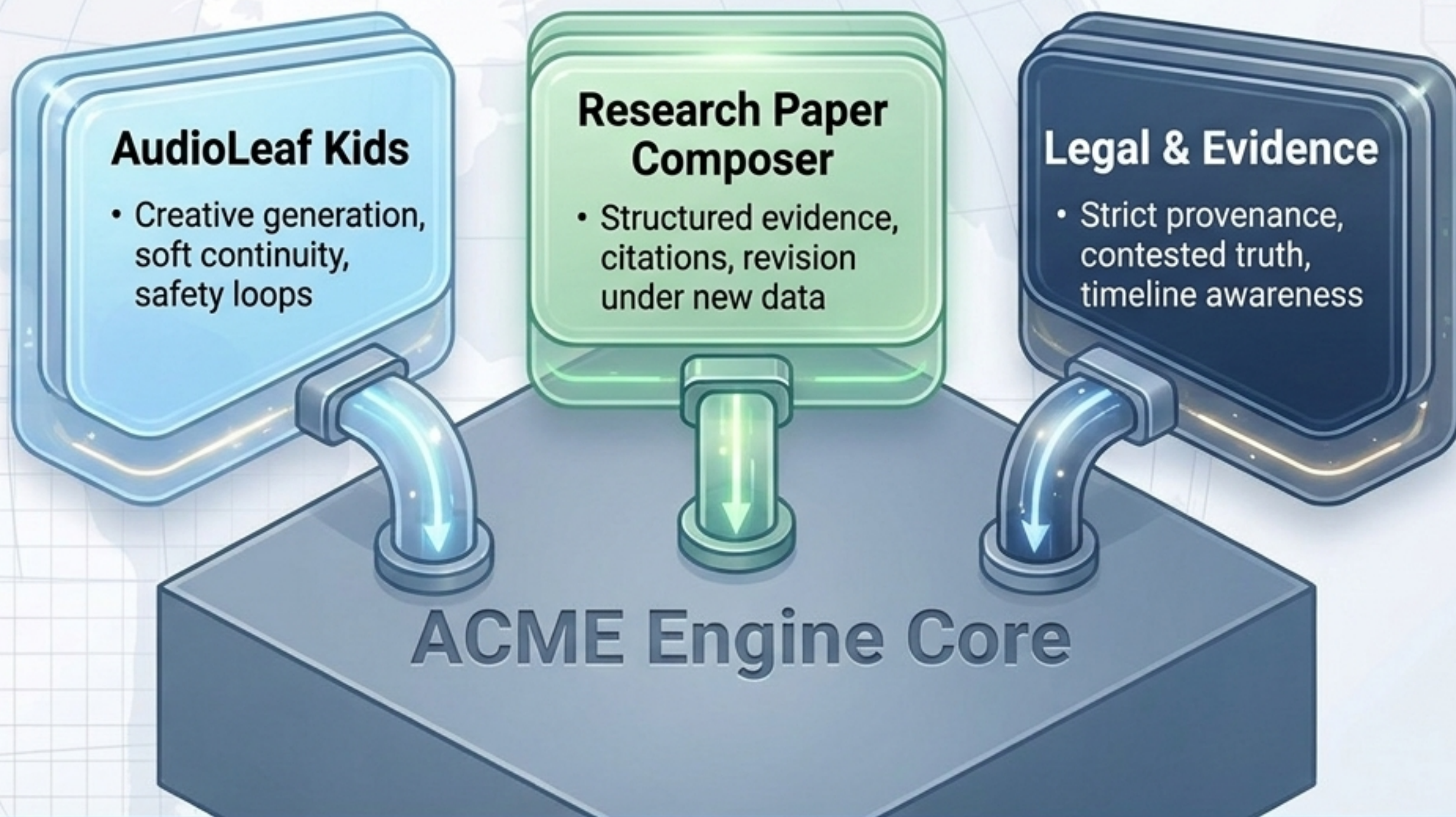
The Core handles the physics of retention. The Domain dictates the laws of meaning.

Durability is proven behavior, not an assumption



An interrupted execution completes from its recorded model call.
Failure is an expectation, handled natively by the revisioned Unit of Work.

The Platform Proof: Three domains, one unchanged core

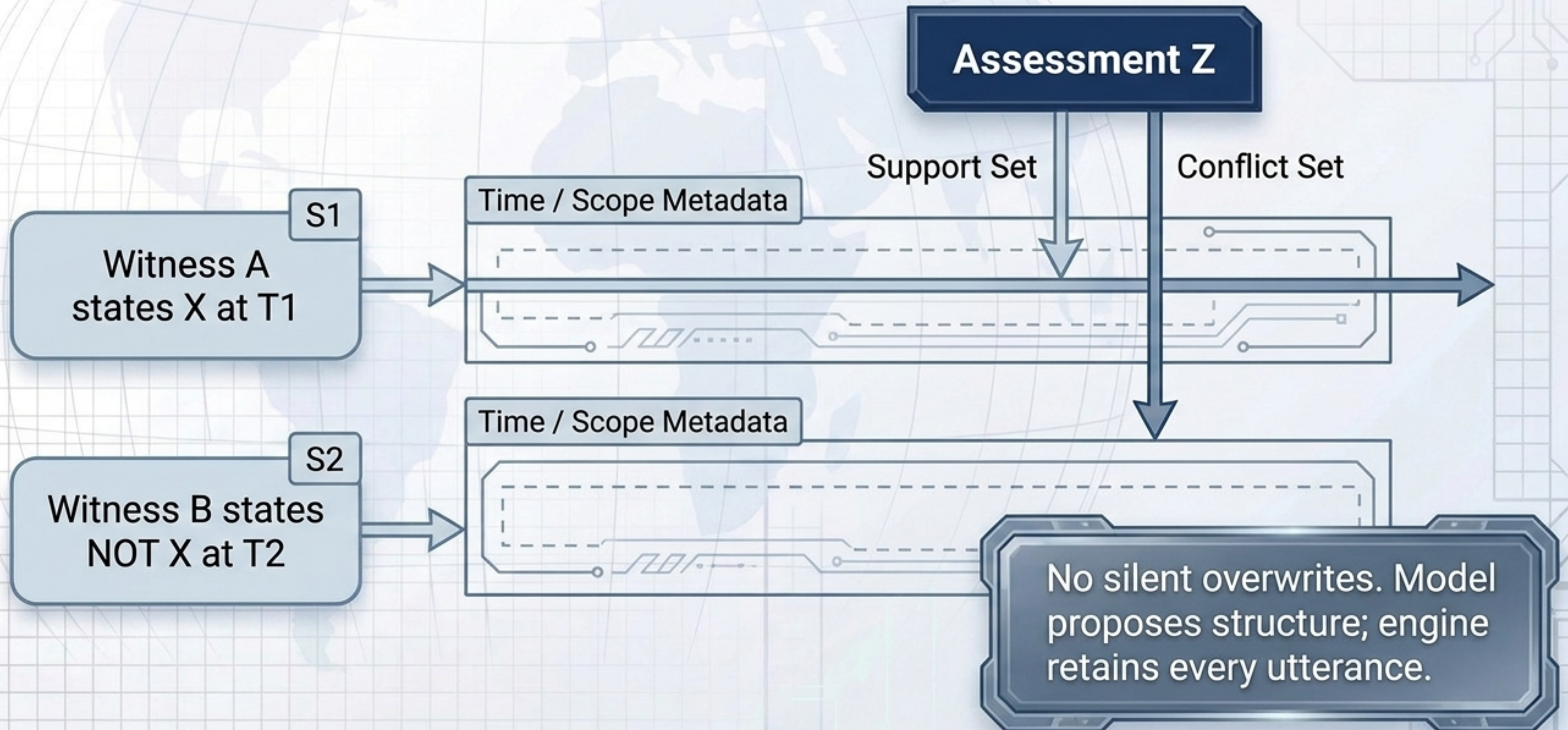


If the core requires domain-specific vocabulary to run these, the architecture has failed. ACME runs all three with zero domain words in the core.

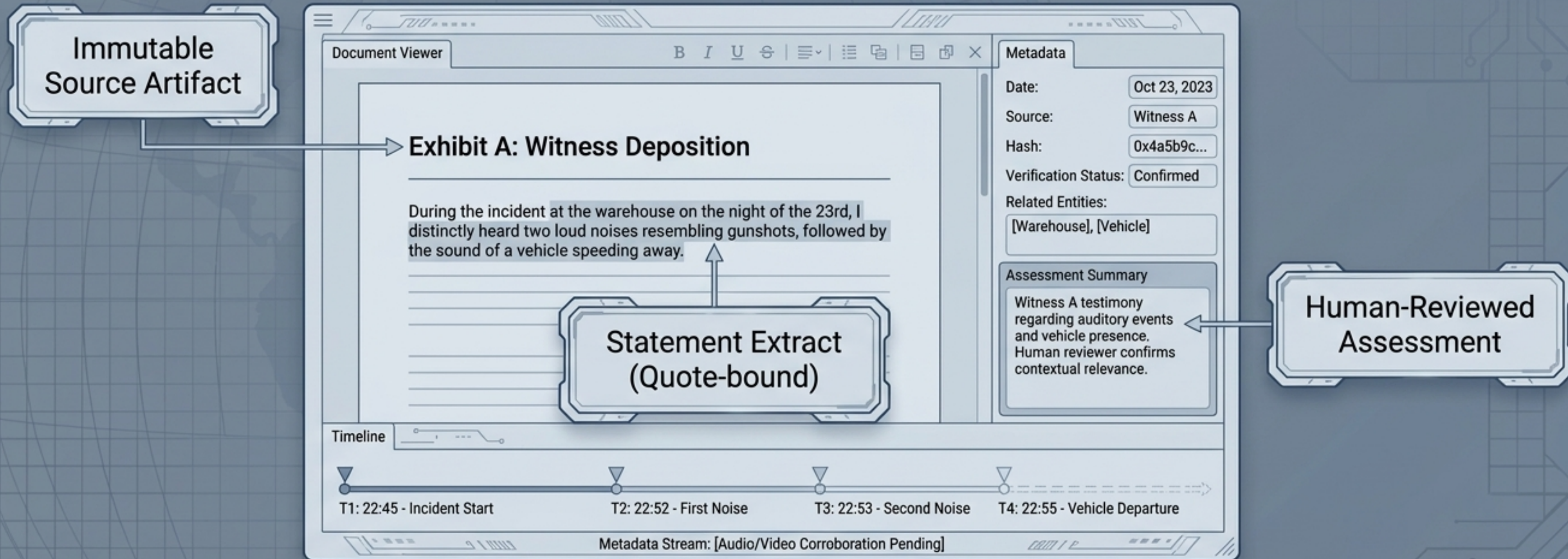
How the same engine handles vastly different pressures

Dimension	AudioLeaf Kids	Research Paper	Legal / Evidence
Provenance	Creative generation, soft continuity	Structured evidence, citations revisiort	Strict provenance, contested truth
Contradiction	Contradiction contraduction	Contested truth prorranem	Strict contest or prvareness
Uncertainty	Uncertainty or in the hast	Uncertainty is uncompocvribility	Uncertainty did not use mist contest
Overwrite Rule	Continuity apply	Never silent; supersede only with stronger support	Never silent; contest or coexist by scope

The hardest AI problem: Contradiction $\neq$ Deletion

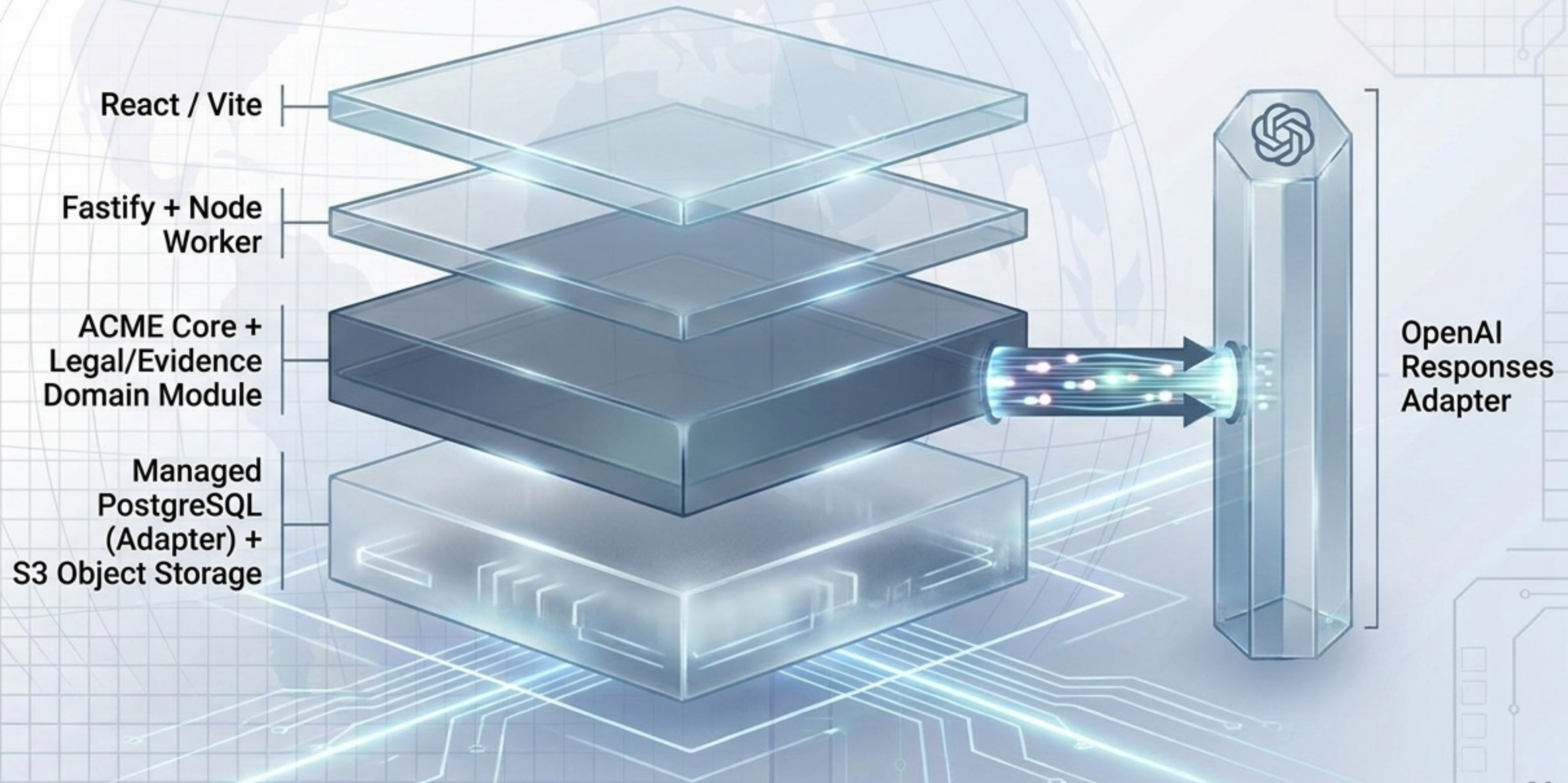


POC #1: The Evidence Integrity Workbench



A corpus-bound, non-adjudicative review product. Explicitly prohibited from generating legal advice, automated guilt determinations, or credibility scoring.

The Workbench Architecture & Stack



Local Verification: The Domain Test UI



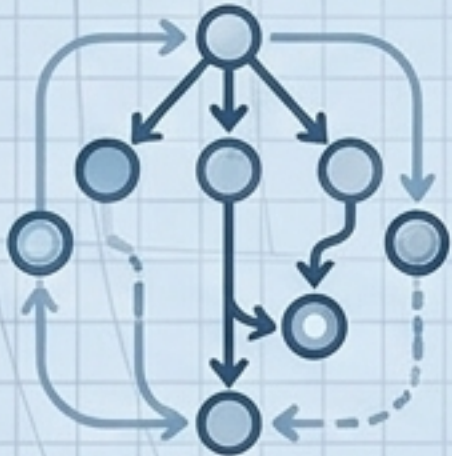
S1 Catalog

Static registries & discovered scenarios



S3 Launch Console

App-owned run records



S6 State Lineage

Ordered revision lineage and continuity



S7 Replay Inspector

Exact engine verdicts (match/different) offline

Before deploying, developers require local, visual proof. An offline, secret-free CI workbench that verifies every transaction without a single provider call.

Separating Pre-Commit Gates from Quality Evaluation



Current Status & Horizon

 Delivered	 Deferred / Horizon
 Domain-neutral core  SQLite ledger & outbox delivery  OpenAI Gateway with schema lowering  Domain Test UI (S1-S10 loopback)  Legal POC defined and accepted	 Production hosting & Managed PostgreSQL  Multi-tenant authorization  KMS / Encryption Key Rotation  Vector retrieval integration

The foundation is verified. Open work remains strictly bounded behind future charters.